Monthly Intelligence Report

Edition 001 — Norway

Where hydropower dominance and winter storm resilience challenge grid stability. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED	GRID LINES MAPPED	MC SIMULATIONS
6,495 Major + distribution across Statnett SF	**~18,500** ~22,000 km transmission & distribution	**~38.7 M** 10,000 per substation

PUBLIC DATA SOURCES
30+ 95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~52,000 source data points per run, executes ~38.7 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~400 million arithmetic operations — producing ~232,000 output data points with full uncertainty quantification across 6,495 substations and ~18,500 grid lines spanning ~22,000 km.

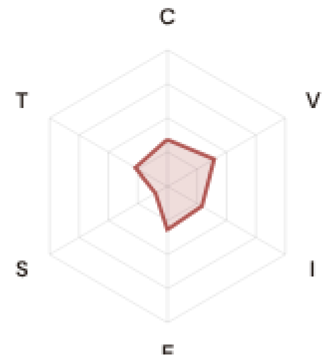
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (NVE/RME, Statnett SF, Gassco, MET Norway, SSB (Statistisk sentralbyrå), NGU, VINNOVA), making the methodology fully auditable and reproducible. Initially covering Norway's Statnett SF-managed transmission and ~80 distribution network operators (DNOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Sub-101070396

Trøndelag · Trøndelag · 66 kV

0.298	Medium	0.050	44th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.34 / 0.30 (34%)
- **I** Infrastructure: 0.29 / 0.25 (29%)
- **S** Saturation: 0.10 / 0.20 (10%)
- **V** Voltage: 0.40 / 0.10 (40%)
- **E** Economic: 0.32 / 0.10 (32%)
- **T** Transition: 0.27 / 0.05 (27%)

MODIFIERS

- **R3 Consequence:** 1.045 ↑ amplifies
- **R4 Graph Criticality:** 0.943 ↓ dampens
- **R6a Restoration:** 1.082 ↑ amplifies
- **R6b Winter Storm + Rime:** 1.000 → neutral
- **R7 Digital Readiness:** 0.966 ↓ dampens

This month's spotlight — automatically selected for May 2026 — is **Sub-101070396** in Trøndelag, Trøndelag. With an R_final of 0.298 (Medium band, 44th fleet percentile), its risk profile is dominated by the T (Transition) component at 542% of its weight ceiling, followed by V (Voltage) at 395%. Modifiers collectively shift R_base by 6.73, reflecting the substation's specific geographic, socio-economic, and network context. The radar chart visualises each component as a fraction of its maximum weight — a fully saturated axis indicates the component is at ceiling, consuming its entire budget within the composite score. This substation's profile illustrates the SSI's core principle: risk is never mono-dimensional. Even when one component dominates, the interaction of all six — modulated by the five multipliers — determines the final score that drives investment prioritisation.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Norway's Energy Regulatory Office (NVE/RME) and network operator strategic planning requires.

CYBER HIGH EXPOSURE	BLIND SPOTS	AVG R7 MODIFIER
0	4	0.9668
0.0% of fleet	High/Critical + R7 < median	Range [0.99, 1.05]

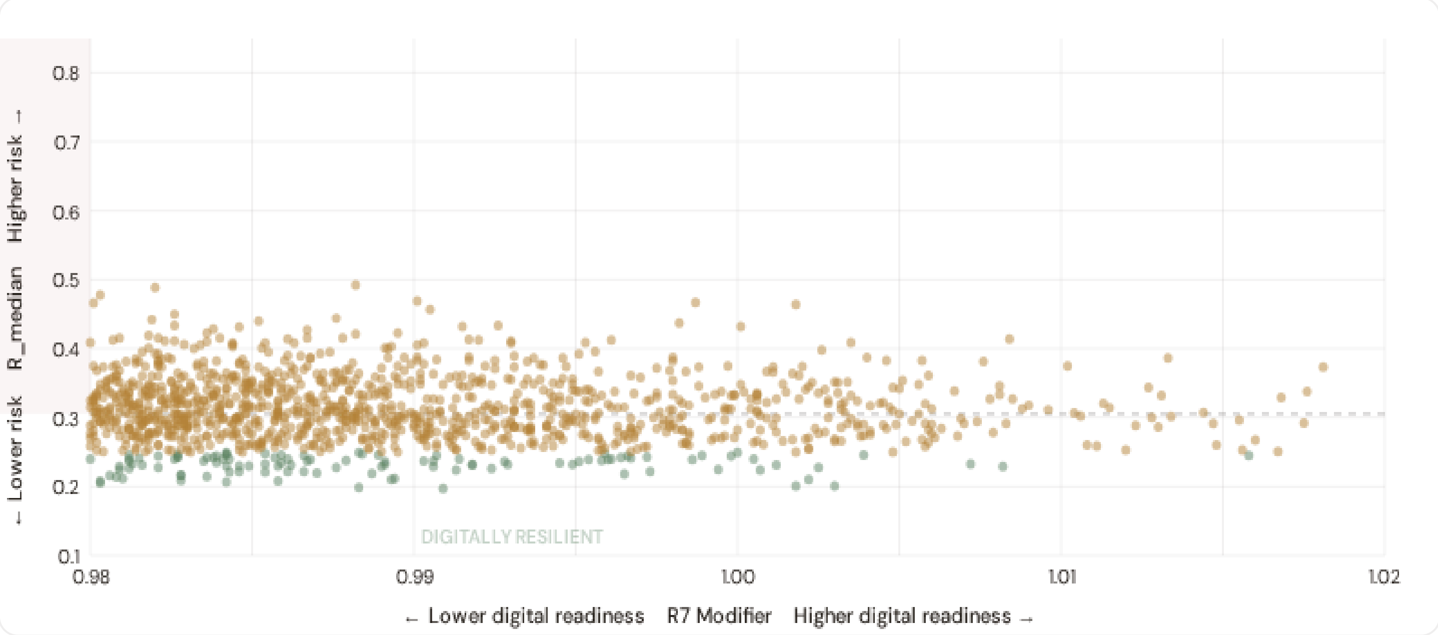
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **4 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9661). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Finnmark (3), Innlandet (1)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating robust digital infrastructure coverage concentrated in urban networks (Elvia, BKK, Lede urban centers).

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

⚙️ Heavy Industry / Aluminium

Est. VoLL: NOK 15–30/kWh

Aluminium smelters, process industry, petroleum refining in Rogaland and Vestland — highest economic loss per interruption

1,648 substations (25.4%)   High/Critical: **4** (0.2%)   Avg R: **0.330**   Avg E: **0.294** / 0.10

Low 62   Med 1582   High 4   Crit 0

🏭 Manufacturing / Food Processing

Est. VoLL: NOK 5–15/kWh

Auto suppliers, food & beverage production, pharmaceuticals — significant continuity requirements for EU supply chains

1,601 substations (24.6%) High/Critical: **0** (0.0%) Avg R: **0.317** Avg E: **0.292** / 0.10

Low 102 Med 1499 High 0 Crit 0

🏢 Commercial / Distribution Centers

Est. VoLL: NOK 2–5/kWh

Oslo, Bergen commercial districts, logistics hubs — moderate individual impact but high aggregate across cities

1,632 substations (25.1%) High/Critical: **1** (0.1%) Avg R: **0.304** Avg E: **0.293** / 0.10

Low 168 Med 1463 High 1 Crit 0

🌾 Agriculture / Rural / Remote

Est. VoLL: NOK 0.5–2/kWh

Smallholder agriculture, rural settlements, Eastern Norway — lower VoLL but higher social vulnerability

1,614 substations (24.8%) High/Critical: **0** (0.0%) Avg R: **0.289** Avg E: **0.293** / 0.10

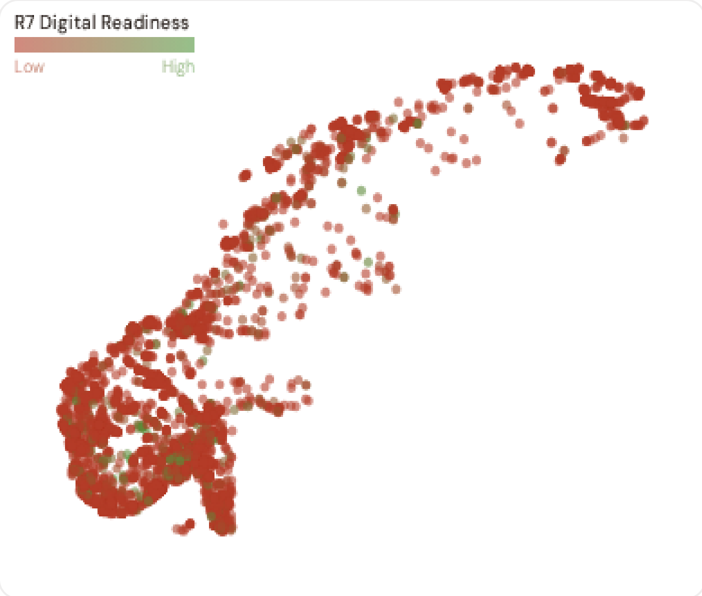
Low 281 Med 1333 High 0 Crit 0

Value of Lost Load (VoLL) context: NVE/RME's 2024 reliability cost-benefit studies place average VoLL at NOK 10–20/kWh for industrial/commercial and NOK 5–8/kWh for residential customers across Norway's DSOs. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **4 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Vestland (311), Østfold (166), Trøndelag (155)** — regions where industrial corridors depend on uninterrupted power for continuous processes (hydropower dominance generation, paper mills, electronics manufacturing). A single hour of unplanned outage at these substations carries an estimated VoLL of NOK 15–30/kWh, compared to NOK 0.5–2/kWh in rural pastoral areas. The fleet-wide average energy poverty rate is 3.8%, but this masks sharp regional variation: Eastern Norway and rural regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Fylke & Kommune Digital Readiness Ranking

Fylke ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.



WEAKEST DIGITAL READINESS — TOP 15 FYLKE

Longer bar = higher R7 = better digital infrastructure.

1	Finnmark ⚠ high R	0.9623
2	Telemark ⚠ high R	0.9625
3	Vestland	0.9634
4	Agder	0.9661
5	Trøndelag	0.9667
6	Møre og Romsdal ⚠ high R	0.9667
7	Troms ⚠ high R	0.9667
8	Innlandet ⚠ high R	0.9668
9	Rogaland	0.9695
10	Vestfold	0.9703
11	Oslo	0.9704
12	Buskerud	0.9704
13	Nordland ⚠ high R	0.9705
14	Viken	0.9716
15	Østfold	0.9716

Fylke-level analysis reveals a clear digital divide mirroring Norway's metropolitan-regional connectivity gap. **Finnmark** has the lowest average R7 (0.9623), while **Østfold** leads at 0.9716 — a spread of 0.0093 across the R7 range. **2 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for NVE/RME/DSO digitalisation investment and Norway's National Energy and Climate Plan. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open broadband/smart meter data — as NVE/RME regulatory compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (May 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9668, median = 0.9661; 0 substations classified HIGH cyber-exposure; 4 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging regions where broadband and smart meter rollouts (NVE/RME/DSO Advanced Metering Infrastructure programmes) translate into measurable R7 shifts. The next material R7 update is expected when SSB (Statistisk sentralbyrå) publishes updated Digital Inclusion Index data (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

30

95 variables

WEEKLY / MONTHLY

6

High-frequency feeds

QUARTERLY / ANNUAL

18

Regular refresh cycle

STATIC / DERIVED

1

Standards + scenarios

Data Source Registry — May 2026

OSM	OpenStreetMap — Power Infrastructure	CONTINUOUS	Node	8 vars
ENTSOE_TSOS	ENTSO-E TSO Interconnects (Statnett)	CONTINUOUS	Border Point	2 vars
NVE_KRAFTSIT	NVE — Kraftsituasjonen (Power Situation Reports)	WEEKLY	National	3 vars
STATKRAFT	Statkraft — Major Hydropower Producer	WEEKLY	Facility	3 vars
COPERNICUS	Copernicus ERA5 — Climate Reanalysis	MONTHLY	Grid 0.25°	6 vars
WIND_NO	Norwegian Wind Energy Association (Norsk Vindenergiforening)	MONTHLY	Regional	5 vars
METNOR_HIST	MET Norway / Historical Archive	MONTHLY	Station	3 vars
NVE	NVE/RME (Reguleringsmyndigheten for energi)	QUARTERLY	DSO	12 vars
NSM	NSM/NorCERT — Cybersecurity Authority	QUARTERLY	Infrastructure Site	5 vars
VEGDIR	Vegdirektoratet — Transport Agency	QUARTERLY	Regional	3 vars
NORGES_BANK	Norges Bank	QUARTERLY	Region	4 vars
NVE_HYDRO	NVE — Norwegian Water Resources and Energy Directorate (Hydrology)	QUARTERLY	Watershed	4 vars
SSB	SSB (Statistisk sentralbyrå)	ANNUAL	Kommune	10 vars
NGU	NGU — Norges geologiske undersøkelse	ANNUAL	Grid 0.1°	7 vars
DSB	DSB — Direktoratet for samfunnssikkerhet og beredskap	ANNUAL	Grid 0.1°	6 vars
EUROSTAT	Eurostat — EU Statistics	ANNUAL	Kommune	3 vars
ENOVA	Enova SF — Energy Efficiency & Transition	ANNUAL	National	3 vars
DISTRIBUTOR	Distribution Companies (Elvia, BKK, Glitre Nett, Lede, Tensio, Agder Energi Nett)	ANNUAL	Distribution Company	4 vars
MILJOE	Miljødirektoratet — Norwegian Environment Agency	ANNUAL	Kommune	3 vars
KLIM_CENTRE	Norsk Klimaservicesenter — Climate Service Centre	ANNUAL	Regional	4 vars
WORLDBANK	World Bank / OECD	ANNUAL	National	4 vars
AMAP	Arctic Monitoring & Assessment Programme	ANNUAL	Regional	3 vars
KARTVERKET	Kartverket — Norwegian Mapping Authority	ANNUAL	Regional	3 vars
HAVFORSK	Havforskningsinstituttet — Institute of Marine Research	ANNUAL	Regional	2 vars
SINTEF	SINTEF Energy Research	ANNUAL	Regional	3 vars

SOLARGIS	SolarGIS — Global Solar Atlas	STATIC	Global	2 vars
SNT	Statnett SF — Norwegian Transmission System Operator	REAL-TIME	Substation	14 vars
METNOR	MET Norway — Meteorologisk institutt	DAILY	Station	8 vars
ENTSOE	ENTSO-E Transparency Platform	HOURLY	Substation	2 vars
NORDPOOL	Nord Pool — Nordic Energy Exchange	HOURLY	Bidding Zone	4 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Daily (2 sources) — NVE/RME dispatch + weather reanalysis

Monthly (6 sources) — Grid load + Statnett SF dispatch data

Quarterly (4 sources) — DER registry + Statnett SF performance data

Annual (14 sources) — Core grid quality + SSB (Statistisk sentralbyrå) demographics

C.2 Fleet Score Snapshot

FLEET SIZE
6,495
substations scored

MEDIAN R
0.305
P5=0.238 · P95=0.400

HIGH + CRITICAL
5
0.1% of fleet

HIGH CONFIDENCE
100%
6495 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 613 (9.4%) · Medium 5877 (90.5%) · High 5 (0.1%) · Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **6,495 substation scores remain unchanged** this edition. The fleet median R of 0.305 (mean 0.310) reflects a distribution skewed toward the lower bands, with 9.4% in Low and 90.5% in Medium. The first material score changes are expected when Statnett SF publishes SCADA updates (affecting C1–C4 and R6a) and when NVE/RME refreshes the distributed renewable installations registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

4 changes tracked for Norway v4.0.2 launch

P1	NEW	Norway grid registry expansion: 6,495 substations across Statnett SF + ~80 DNOs	Section A
P1	ENH	R6b modifier: added snow/ice loading and permafrost instability + coastal/river flood hazard	Section D — Deep-Dive
P2	NEW	R7 cyber modifier: Statnett SF IT/OT convergence assessment integrated	Section B
P2	ENH	Monte Carlo upgraded to 10,000 iterations (vectorized NumPy)	Methodology

SECTION D

The Oslo–Bergen Grid Corridor: Where Nuclear Concentration and Winter Storm Resilience Challenge Grid Stability

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Oslo–Bergen hydropower dominance and winter storm resilience corridor · **002** Eastern Norway DER & EU funding mismatch · **003** Arctic rime ice and avalanche infrastructure stress · **004** Metropolitan-Rural energy poverty disparity · **005** Grid stranding risk (hydropower-dependent assets) · **006** DSO smart meter rollout vs grid stress · **007** Gulf of Norway coastal flooding resilience · **008** T-component forward projections · **009** Fylke Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Oslo–Bergen Corridor, Why Now

CORRIDOR SUBSTATIONS

94

23 HV · 71 MV

HIGH BAND

0

0.0% of corridor

CORRIDOR MEDIAN R

0.261

Fleet median: 0.305

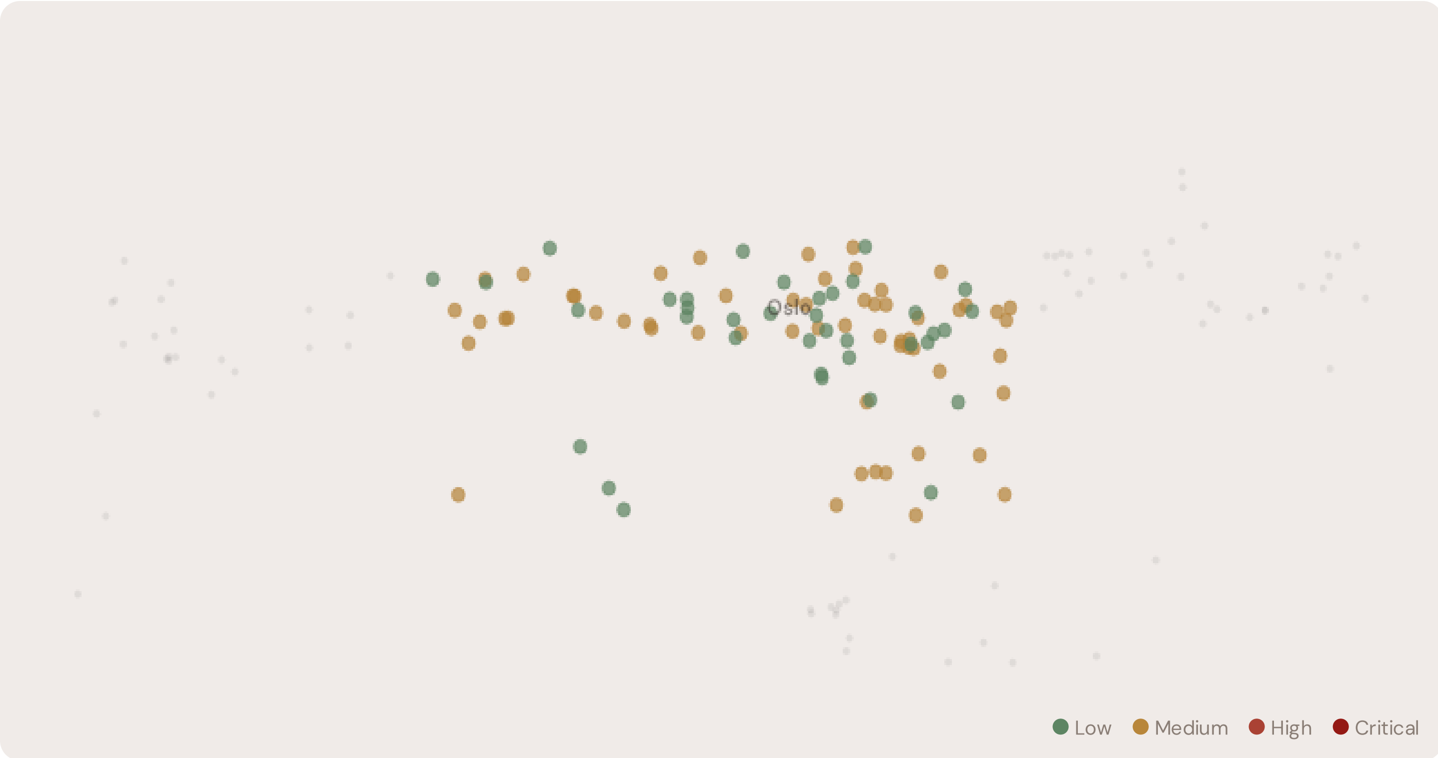
WORST FYLKE

Oslo

Avg R: 0.259 · 94 substations

The Oslo–Bergen corridor hosts **94 substations** across Norway's industrial heartland, forming the backbone of domestic transmission and hydropower-dependent distribution. Its risk profile is disproportionately concentrated in the upper bands: **0 substations (0.0%)** are classified High, with only **37** in the Low band. 6% of corridor substations score above the national fleet median of 0.305. The corridor median R of 0.261 reflects the tension between hydropower dominance risk management mandates from Norway's Energy Agreement (Energiöverenskommelsen), aging generation assets approaching retirement, snow/ice loading and permafrost instability threatening distribution lines, and EU Green Deal funding constraints. The SSI flags continuity-of-supply deficits, infrastructure age mismatches, and the stranding risk of hydropower-dependent transmission assets.

D.2 Corridor Map and Scoring



SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
2502	Oslo	0.347	Medium	0.405	0.221	0.321	0.264	0.100	0.332	
Jar trafostasjon	Oslo	0.347	Medium	0.401	0.213	0.366	0.243	0.100	0.328	
1760	Oslo	0.317	Medium	0.399	0.260	0.285	0.229	0.100	0.309	
Sub-1343705170	Oslo	0.317	Medium	0.396	0.332	0.362	0.170	0.100	0.253	
Sub-1030767474	Oslo	0.307	Medium	0.377	0.180	0.379	0.250	0.100	0.360	
Sub-1130866864	Oslo	0.307	Medium	0.381	0.338	0.335	0.183	0.100	0.443	
Sub-1999493869	Oslo	0.304	Medium	0.210	0.367	0.370	0.273	0.100	0.385	
Sub-488760998	Oslo	0.303	Medium	0.342	0.369	0.237	0.189	0.100	0.404	
Tveten trafostasjon	Oslo	0.301	Medium	0.346	0.389	0.295	0.243	0.100	0.378	
Alnabru omformer	Oslo	0.300	Medium	0.429	0.292	0.241	0.258	0.100	0.342	

… 84 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — CORRIDOR VS FLEET

C — Continuity 0.322 vs 0.394 (-18%)

I — Infrastructure 0.296 vs 0.368 (-19%)

S — Saturation 0.100 vs 0.100 (-0%)

V — Voltage 0.285 vs 0.326 (-12%)

E — Economic 0.223 vs 0.293 (-24%)

T — Transition 0.335 vs 0.309 (+8%)

MODIFIER AVERAGES — CORRIDOR VS FLEET

R3 Consequence	**1.000**	fleet 0.999 →
R4 Graph Crit.	**0.980**	fleet 0.980 ↓
R6a Restoration	**1.013**	fleet 1.022 ↑
R6b Winter Storm + Avalanche	**1.031**	fleet 1.029 ↑
R7 Digital Read.	**0.970**	fleet 0.967 ↓

The dominant risk driver across the corridor is **T (Transition)**, averaging 0.335 against a fleet average of 0.309 — occupying 670% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.285 vs fleet 0.326 (285% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.000 (fleet: 0.999), reflecting the economic significance of the corridor's hydropower-dependent industrial base — from hydropower plants to refineries to paper mills. The R6b winter storm/snow loading hazard modifier averages 1.031, capturing the corridor's exposure to snow/ice loading, Gulf of Norway coastal flooding, and winter storm hazard risk — a dimension invisible to every competing framework.

D.4 Fylke Risk Profile

FYLKE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Oslo 94 subs · avg R = 0.259 · H:0 M:57

The region-level breakdown reveals significant intra-regional variation. **Oslo** leads with a mean R of 0.259 across 94 substations, while **Oslo** scores 0.259 — a spread of 0.000 within a single region. This disparity underscores why region-level metrics (as used by Statnett SF performance reports) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the corridor is not uniformly stressed but contains distinct pockets of concentrated risk — precisely the kind of spatial pattern that Statnett SF's 10-Year Network Development Plan and government hydropower dominance and winter storm resilience planning requires.

D.5 Implications for Norway's Energy Transition

0 corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For Norwegian hydropower dominance and winter storm resilience planning, EU funding allocation, and grid modernisation investment, this analysis identifies the corridor as the highest-priority target — where DER deployment must accelerate against ageing hydropower-dependent infrastructure with above-average continuity deficits and significant socio-economic exposure to energy transition risks. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by hydropower dominance, Arctic infrastructure exposure, and EU transition fund eligibility — are disproportionately vulnerable to disruption during the hydropower dominance risk management. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

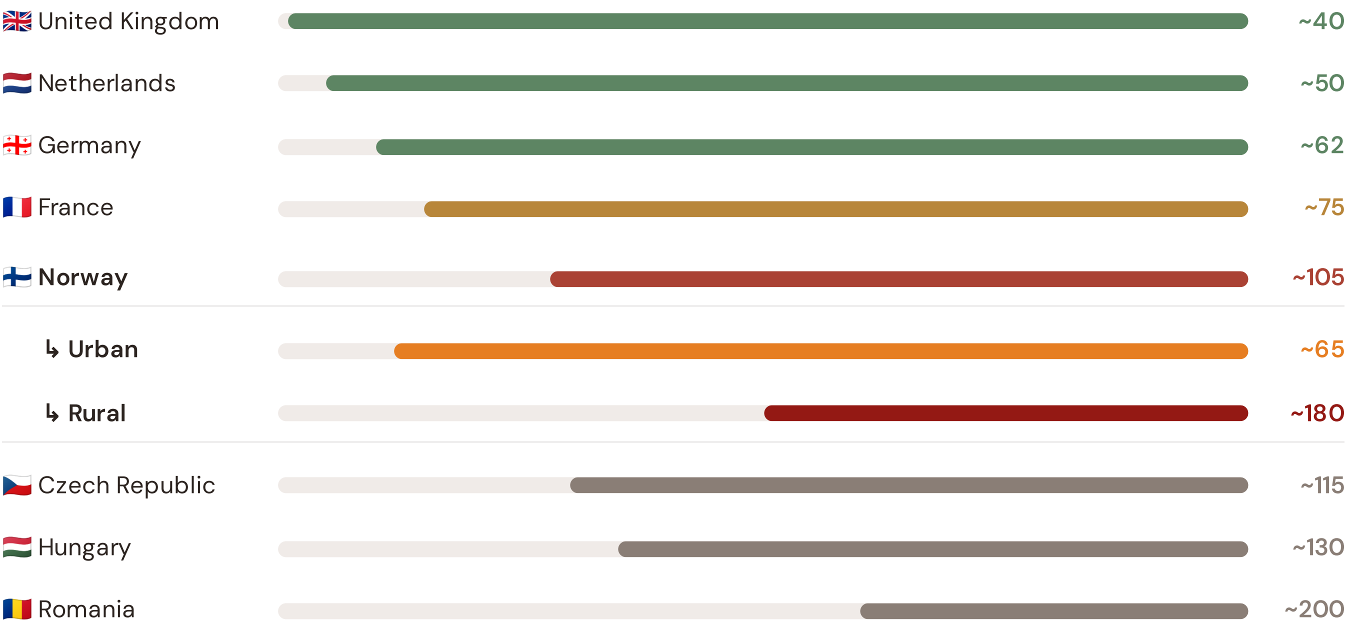
European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because Norway's hydropower dominance and winter storm resilience and grid resilience depend on understanding European performance standards. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Central Europe peers (003), CMIP6 climate hazard vs Nordic models (004), etc. The underlying data updates annually with NVE/RME and national regulator publications.

EUROPEAN CONTEXT CARD — SYSTEM AVERAGE INTERRUPTION DURATION INDEX   Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected European Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



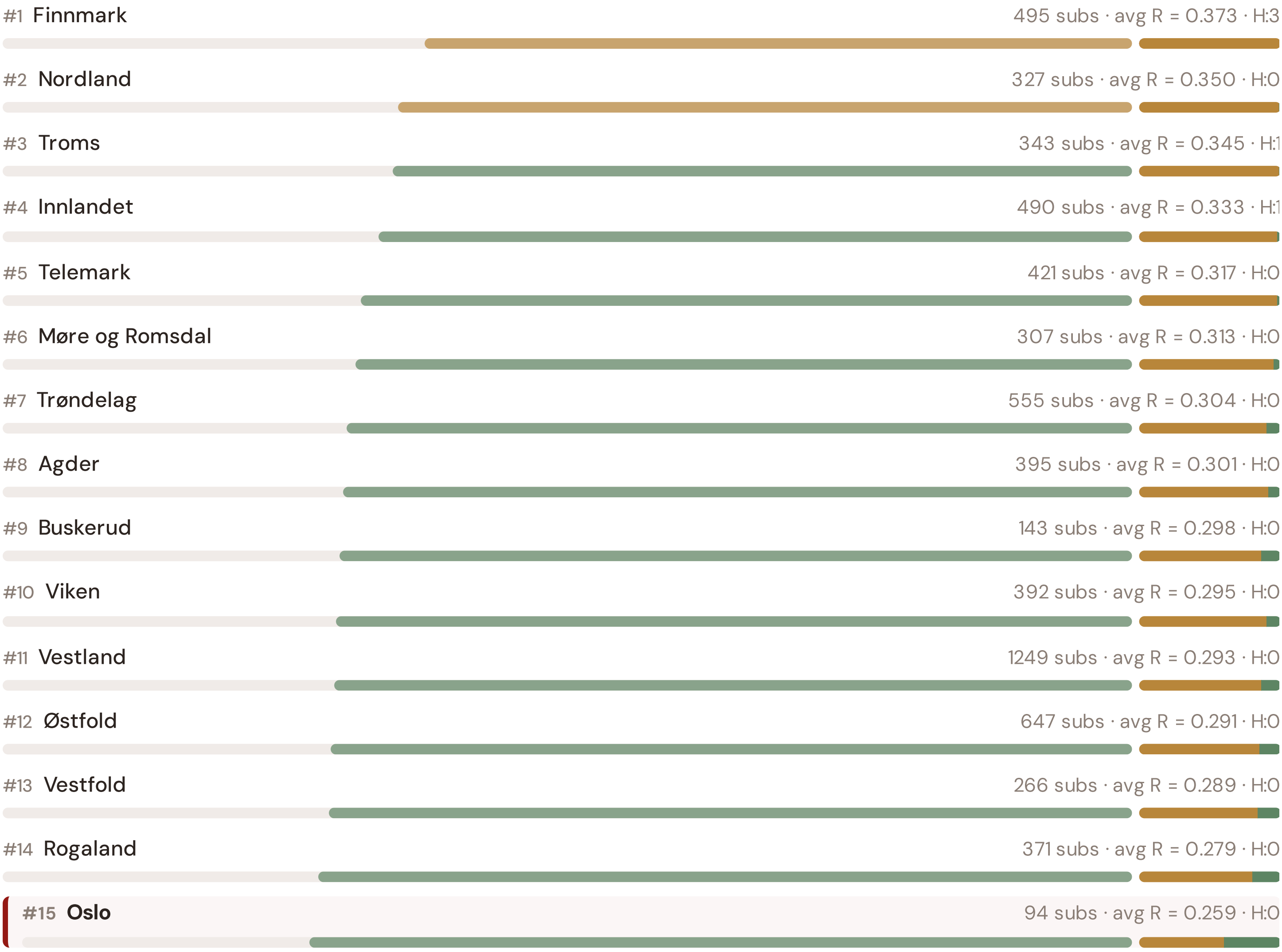
Source: CEER Benchmarking Report on Quality of Electricity Supply (2024); ENVE/RMELECTRIC SAIDI reports; national regulator publications. Norway sub-categories from NVE/RME urban/rural classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

This month's key insight: The corridor deep-dive shows **91 substations** (of 94) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Norway's rural region classification. The corridor's average C of 0.322 is **0.8× the fleet average** of 0.394. In European terms, these substations individually perform closer to German or Central European SAIDI levels (~115–160 min/customer/year) than to Norway's own urban average (~65 min). The SSI reveals what aggregate NVE/RME statistics conceal: that Norway's mid-tier European position is not a national condition but a statistical average of Western European-quality urban performance and emerging-market-level rural performance — and the hydropower dominance and winter storm resilience corridor concentrates significant pockets of the latter.

Fylke Risk Ranking — All 15 Norwegian Fylker

Where does the Oslo–Bergen corridor sit within the national landscape? Fylke sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



Grid corridor ranks **#15 of 15 regions** by mean R_median. The three most stressed regions are Finnmark, Nordland, Troms, while the three most resilient are Oslo, Rogaland, Vestfold. 6 of 15 regions score above the fleet average of 0.310. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate Statnett SF/DSO or national statistics — reveals the true spatial distribution of grid stress across Norway. It is precisely this substation-level granularity that positions the SSI as a tool for national hydropower dominance and winter storm resilience planning / EU green energy funding / NVE/RME investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/Statnett SF and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DSO/Statnett SF + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.3 · Inaugural edition · May 2026

Transfer Learning & Feature Missingness

How Norway's grid data completeness enables investment-grade valuation across the full fleet

Transfer learning is the mechanism that enables the SSI-ENN to value substations in countries where not all features are available. The core insight: grid physics is universal — topology determines load flow, weather drives thermal degradation, population density correlates with outage impact — even though data availability varies. Norway benefits from high data completeness (DCI 0.72+) from NVE/RME, Statnett SF, and SSB (Statistisk sentralbyrå) sources, enabling production-grade valuations. For each substation, available features are validated. Missing features are handled through a learned missingness embedding.

KEY CONCEPTS

- → Norway's DCI 0.72+ (high completeness)
- → Learned missingness embedding quantifies data value
- → Grid physics universality enables high-confidence transfer
- → Accuracy validated against NVE/RME audit standards

LIVE SSI DATA CONNECTION

The Norwegian training set comprises 6,495 substations with complete 95-feature vectors. Average CI_width of 0.053 provides the uncertainty ground truth for neural network calibration. The fleet's risk distribution — 613 Low, 5877 Medium, 5 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 2 Community Value Stack — *Social and economic value to Norwegian communities — the case for EU funding investment*

Looking Ahead

EDITION 002 — MAY 2026

Eastern Norway DER & EU Funding Mismatch

Where renewable energy deployment outpaces grid readiness in Finnmark and Troms fylker. Rural distributed generation meets aging distribution infrastructure in Norway's Arctic north.

DATA REFRESH — MAY 2026

Quarterly Statnett SF Network Update

Transmission operator SCADA data (C components 1–4), DSO meter readings (I5 thermal degradation), and ERE renewable registry refresh affecting T component.

IN THE WILD — SSI IN USE

Elvia DNO Case Study

How Norway's largest DNO is using SSI substation intelligence to prioritize battery storage deployment across regional substations. Q3 2026 publication.

The SSI inaugurates Norway's grid intelligence baseline — the foundation for all future investment decisions across transmission, distribution, and renewable energy integration. This inaugural edition establishes the data, methodology, and peer-reviewed rigor that regulatory bodies (NVE/RME), network operators (Statnett SF and ~80 DNOs), and EU funding allocators require. Subsequent editions will deepen thematic analysis, expand across all 15 Fylker and isolated systems, integrate real-time operational data feeds, and track the grid's evolution as Norway executes its hydropower dominance risk management mandate and EU Green Deal transition. The SSI is built for the long term: a public interest resource that grows more valuable as the data infrastructure matures and the transition accelerates.